

Srushti Raste

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Education

Boston University

Master of Science in Robotics & Autonomous Systems

Boston, USA

Sept 2023 – Jan 2025

University of Mumbai, KJSCE

Bachelor of Technology in Electronics Engineering

Mumbai, India

Aug 2019 – May 2023

Research Experience

Research Associate, Soft Robots Lab, Wesleyan University

March 2025 – Present

Advisor: Professor Sonia Roberts

- Lizard-inspired locomotion on granular media, supported by NSF under Award 2334860
 - Led the upgrade of a legacy legged jumping robot by re-designing electronics, sizing motors and re-engineering the power system.
 - Integrated high-torque direct-drive motors (T-Motor U8 Lite), ODrive Pro controllers, Teensy 4.1 microcontroller, Magnetic encoder and faster communication protocols.
 - Iteratively designed and fabricated aluminum mounts to keep the robot's weight at 2kg despite heavier hardware, using a ProtoMAX waterjet cutting machine.
 - Machined a fixture for the FUTEK load cell on a Tormach CNC mill for vertical alignment.
 - Developed a torsional spring control framework with motor position control and implemented PD tuning to enable controlled jumps on acrylic beads (sand analog).
 - Collecting force measurements, jump height, and battery efficiency data for different foot shapes using DMD 465 amplifier, LCM300 Loadcell, Pololu current sensor and string pot.
 - Documented and open-sourced full workflow including fabrication, assembly, testing, tuning, and troubleshooting to enable reproducibility for other robotics researchers.
- Soft strain and touch sensor, supported by NSF under Award 2347708
 - Developed an assistive glove incorporating a novel resistive strain sensor to trigger soft pneumatic actuators to demonstrate the sensor's potential in grasping.

Graduate Student Researcher, Morphable Biorobotics Lab

Sept 2024 – Dec 2024

Advisor: Dr. Leah Gaeta, Professor Tommaso Ranzani

Boston University

- Designed a flexible PCB for a wearable haptic glove providing tactile feedback via stiffening of metal-jamming sheets actuated by Electropermanent magnets.
- Performed schematic design, component selection, track routing, signal flow, and soldering.
- Facilitated experiments and conducted user studies to evaluate the glove's performance.

Research Assistant, Mechanics of Slender Structures Lab

June 2024 – Aug 2024

Advisor: Professor Douglas Holmes

Boston University

- Designed and fabricated Yoshimura-patterned origami soft actuators using Laser Cutting, exploring multiple bellow geometries and materials for pneumatic extension and bending.
- Conducted constant curvature and grip strength tests to quantify bending and force output.

Publications

- L. T. Gaeta, V. T. Vo, S. Lee, **S. Raste**, M. Venkatesam, J. Rogatinsky, M. D. Albayrak, T. Ranzani ”Jamming Metal Sheets Using Electropermanent Magnets for Stiffness Modulation,” *IEEE Robotics and Automation Letters*, 2025. <https://doi.org/10.1109/LRA.2025.3579246> (peer-reviewed published)
- ”Low cost, easily manufactured, highly flexible strain and touch sensitive fiber for robotics applications” **under review for ICRA, 2026.**

Technical Skills

- **Robotics:** Kinematics, Dynamics, Control, Embedded Systems, Sensor & Actuators
- **Programming:** Embedded C, C++, MATLAB, Python
- **CAD & Circuit Design:** Fusion 360, Altium Designer, Eagle, LTspice, EasyEDA
- **Fabrication:** Tormach CNC Mill, 3D Printing, ProtoMax WaterJetCutter, Laser Cutting, CNC Router, Soldering, Drill Press
- **ML:** Reinforcement Learning, Imitation Learning, TensorFlow, PyTorch, Stable-Baselines3, OpenAI Gym
- **Simulation & Analysis:** MuJoCo, CARLA simulator, Abaqus FEA, Simulink

Mentorship

- Wesleyan University Sept 2025 – Present
 - Mentoring a Wesleyan CS undergraduate student in conducting experiments with the robot for different foot shapes, guiding experimental setup, testing, and data collection.
- Glastonbury GOATS, Smith Middle School, CT Oct 2025 – Nov 2025
 - Mentoring a team of 7 students participating in the First Lego League competition, guiding their innovation project on a smart shovel to assist archaeologists using Arduino and sensor integration.

Projects

- Object Manipulation in Musculoskeletal Simulations Oct 2024 - Dec 2024
 - Trained a Musculoskeletal Arm Model in MuJoCo simulator and MyoSuite library using PPO algorithm to perform hand manipulation tasks.
 - Optimized the training by a custom reward function and hyperparameter tuning & analyzed loss curves.
- Soft Robotic Snake Limb March 2024 - May 2024
 - Iteratively designed and fabricated a soft limb comprising three double PneuNet actuators using DragonSkin 10, along with a granular jamming gripper for grasping.
 - Conducted finite element analysis in Abaqus and mechanical characterization for actuator performance and optimization.

Activities & Outreach

- **KJSCE Robocon** (Nov 2020-Dec 2021) : Participated in the National DD Robocon 2021 as a member of the KJSCE Robocon Mega Project team competition; contributed to robot circuit design, wiring, and troubleshooting.
- Conducted a two-day workshop, mentoring 20 KJSCE undergraduate students on robotics basics and guiding them in building line-following and Wi-Fi-controlled robots using ESP32.
- **Emfinity KJSCE, Math Club**(Jan 2020-May 2020): Conducted engaging workshops to enhance students' understanding of engineering mathematics and its practical applications.
- Volunteered with the Engineering Mathematics Teachers Society, India, for eMAPCo registration (2020), supporting STEM outreach programs in engineering mathematics.